Topological Memory.

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Introduction

Today:

- Proof.
  1. Walktrough the math
  2. A
  3. B
- New experimental results.
- Answering the code teleportation.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$-depolarized channel:

- There exists function $M : (0,1) \to (0,1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p$.
- There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n,k)-Toric, then both the swift rule-and the flip upon majority are local decoders.

Storing Logical State

Logical State

For a random variable X we denote by:

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p-depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p-depolarized channel followed by an application of local decoder D.

The Theorem

Theorem

There is $p_{\text{th}} \in (0,1)$ such that for any $p < p_{\text{th}}$ and any bounded function f with a supremum $|f|$ it follows:

$$\mathbf{E}_{\sim q^{(t)}}[f] \leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\mathbf{E}_{\sim 2p(1-p)}[f \circ D] \leq \mathbf{E}_{\sim p}[f \circ D] + |f|e^{-c\sqrt{n}}$$

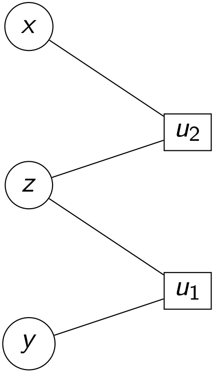
Theorem Proof

By induction on t.

$$
\begin{aligned}
\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}\left[\mathbf{E}_{\sim p}[f \circ D]\right] \\
&\leq \mathbf{E}_{\sim p}\left[\mathbf{E}_{\sim p}[f \circ D]\right] + (t-1)|\mathbf{E}_{p}[f \circ D]|e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\
&\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}
\end{aligned}
$$

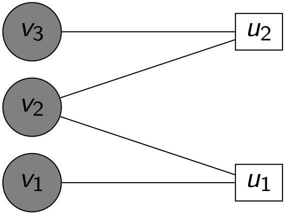
When in the last transition we used the lemma.

Dependence

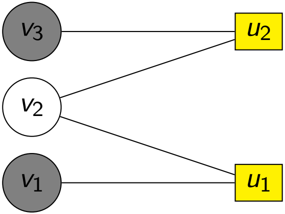


$$
\begin{aligned}
&\mathbf{Pr}\left[(x, y) \in e \middle| z \text{ is not flipped}\right] \\
&= \mathbf{Pr}\left[x \in e \middle| z \text{ is not flipped}\right] \mathbf{Pr}\left[y \in e \middle| z \text{ is not flipped}\right] \\
&\leq \mathbf{Pr}\left[x \in e\right] \mathbf{Pr}\left[y \in e\right]
\end{aligned}
$$

v_2 is flipped.



v_2 is not faulty.



If the majority of bits in the local view of v_2 are flipped, then after decoding, v_2 will be also (remain) flipped.

Lemma Proof

We are going to use the following inequality:

Theorem 1 (McDiarmid's inequality). *Let $X_1, \ldots, X_n$ be independent random variables, where X_i has range $\mathcal{X}_i$. Let $f\colon \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \mathbb{R}$ be any function with the $(c_1, \ldots, c_n)$-*bounded differences property*: for every $i = 1, \ldots, n$ and every $(x_1, \ldots, x_n), (x'_1, \ldots, x'_n) \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ that differ only in the i-th corodinate ($x_j = x'_j$ for all $j \neq i$), we have*

$$|f(x_1, \ldots, x_n) - f(x'_1, \ldots, x'_n)| \leq c_i.$$

For any $t > 0$,

$$\Pr(f(X_1, \ldots, X_n) - \mathbb{E}[f(X_1, \ldots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{n} c_i^2}\right).$$

Lemma Proof